

Bharat Kharpuse

Research Fellow

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Profile Summary

M.Sc. in Physics with a specialization in Nuclear Physics, currently working in medical physics under the FOOT collaboration, contributing to the improvement of particle beam therapy through experimental measurements of nuclear fragmentation cross sections. Experienced in detector calibration (Calorimeter), data acquisition systems, and data analysis using ROOT and Python. Additionally, familiar with Monte Carlo simulations using GEANT4. Strong interest in applying nuclear science to medical and space radiation research.

Research Interests

Medical Physics, Experimental Nuclear Physics, Gamma-ray Spectroscopy, Nuclear Reaction Dynamics,

Education

- 2019–2021 **M.Sc. in Physics (Nuclear Physics)**, IEST, Shibpur, West Bengal, India
CGPA: 7.47 / 10
- 2016–2019 **B.Sc. in Physics**, Devi Ahilya Vishwavidyalaya (DAVV), Indore, MP, India
Percentage: 68.02%

Research Experience

- 04/2025– **Post Graduate Grant**, INFN-Torino, Supervisor: Piergiorgio Cerello
- 03/2026
 - Contributing to the FOOT collaboration with a focus on BGO calorimeter calibration and neutron detection techniques for secondary fragment analysis in ion therapy.
 - Involved in data analysis from experimental campaigns at GSI (Germany) and CNAO (Italy).
- 01/2024– **Junior Research Fellow**, IIT Roorkee, Supervisor: Prof. Moumita Maiti
- 02/2025
 - Working on fusion and breakup reaction dynamics of heavy projectiles.
- 09/2022– **Junior Research Fellow**, Visva-Bharati University, Santiniketan, Supervisor: Dr. Budhhadev Mukherjee
- 01/2024
 - Gamma spectroscopy and lifetime measurement in ^{69}Ge .
 - Participated in INGA Campaign 2023 (VECC, Kolkata).
 - Guided a master student on his master's thesis, titled as "Empirical residual neutron-proton Interactions in nuclei"
- 2021 **M.Sc. Thesis**, IEST, Shibpur, Supervisor: Prof. S.S. Sarkar
- Worked on shell model calculations using one- and two-body operators in finite orbital space.
- 2020 **M.Sc. Term Paper**, IEST, Shibpur, Supervisor: Prof. S.S. Sarkar
- Studied empirical neutron-proton interactions in $N = Z$ nuclei.

Conference Publications *[full list]*

1. **Bharat Kharpuse** and Moumita Maiti, “Analysis of residual cross section from ^{11}B -induced reaction on Zr”, *DAE Symp. Nucl. Phys.* **68** (2025) 399-400.
2. **B. Kharpuse** et al., “In-beam γ -ray spectroscopy of ^{69}Ge ”, *DAE Symp. Nucl. Phys.* **67** (2024) 99-100.
3. **B. Kharpuse** et al., “Shell Model Description of Octupole Correlation in ^{66}Zn and $^{66,68}\text{Ge}$ ”, IC RTP-2023, Abstract Accepted.

Technical Skills

Detector Systems & Experiments	○ Calibration and response studies of BGO Calorimeter (INFN-Torino, FOOT Collaboration) ○ Participated in the setup of Indian National Gamma Array (INGA-2023) campaign (@VECC-Kolkata) using HPGe Clover detectors and PIXI-16 DAQ. ○ Hands-on experience with Scintillator-based setups, SCA/MCA systems, and Compton scattering experiments
Data Analysis Tools	○ ROOT and Python for calibration, peak fitting, energy response and visualization (@INFN-Torino) ○ GENIE 2K for gamma-ray spectral analysis (@IIT-Roorkee) ○ RADWARE, INGASORT and TV for level scheme and coincidence matrix analysis (@Visva-Bharati, Santiniketan) ○ Origin, L^AT_EX for scientific visualization and reporting
Simulations	Familiar with Monte Carlo simulations using GEANT4 for modeling detector systems and particle interactions
Theoretical Tools	Experience with nuclear shell model codes: NuShellX@MSU , KSHELL

Achievements

2024	Madhya Pradesh State Eligibility Test – Qualified
2023 & 2022	Graduate Aptitude Test In Engineering (GATE), Organised by the Indian Institute of Technology
2022	Maharashtra State Eligibility Test – Qualified
2019	Joint Admission Test for Master’s (JAM) -2019 (Rank- 1476)
2019	Among 1% State Toppers – National Graduate Physics Examination (NGPE - 2019), Organised by IAPT
2012	National Means-cum-Merit Scholarship – Recognition of outstanding academic performance in secondary school

Conference Presentations

- 2024 68th DAE Symposium on Nuclear Physics, IIT-Roorkee - **Oral Presentation**.

2023 67th DAE Symposium on Nuclear Physics, IIT-Indore – **Poster Presentation.**
2023 ICRTTP Conference, DAVV Indore – **Poster Presentation.**

Workshops & Schools

16/10/2023– School on Nuclear Models for Structure Studies, IUAC Delhi – Theoretical Training.
22/10/2023
06/2023– Indian National Gamma Array (INGA) Campaign, VECC Kolkata – Experimental Participation.
07/2023
22/02/2023– School on Data Acquisition and Analysis, DAA-2023, IUAC Delhi – ROOT &
24/02/2023 Data Analysis.

Languages

English Fluent
Hindi Fluent

Declaration

I hereby declare that the information provided above is true to the best of my knowledge and belief.

Place: Torino, Italy
Date: August 3, 2025

Bharat Kharpuse